

Midterm exam: April 13

Runge-Kutta Methods

$$u'(t) = f(u, t) \quad u: \mathbb{R} \rightarrow \mathbb{R}^m$$

$$u(0) = \eta$$

RK:

$$Y_i = U^n + k \sum_{j=1}^s a_{ij} f(Y_j, t_n + c_j h)$$

$$U^{n+1} = U^n + k \sum_{j=1}^s b_j f(Y_j, t_n + c_j h)$$

Butcher Tableau

c	$A^{s \times s}$
	b^T

c : abscissas
 b : weights

Examples:

0	0	0
1	1	0
	$\frac{1}{2}$	$\frac{1}{2}$

$$Y_1 = U^n$$

$$Y_2 = U^n + k f(Y_1)$$

$$U^{n+1} = U^n + \frac{k}{2} (f(Y_1) + f(Y_2))$$

$$\tau = O(k^2)$$

Explicit

Kutta's 4th order method:

0	0	0	0	0
$\frac{1}{2}$	$\frac{1}{2}$	0	0	0
$\frac{1}{2}$	0	$\frac{1}{2}$	0	0
1	0	0	1	0
	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{6}$

$$\tau = O(k^4)$$

We always have:

$$C_i = \sum_j a_{ij}$$

Gauss-Legendre

$$\gamma = \frac{\sqrt{3}}{6}$$

$\frac{1}{2} - \gamma$	$\frac{1}{4}$	$\frac{1}{4} - \gamma$
$\frac{1}{2} + \gamma$	$\frac{1}{4} + \gamma$	$\frac{1}{4}$
	$\frac{1}{2}$	$\frac{1}{2}$

$$\tau = O(k^4)$$

Implicit

Need to solve a system of $s \times m$ eqns.

Classes of RK methods:

- Fully implicit
(A is a full matrix)
(solve system of $s \times m$ eqns.)
- Explicit: A is strictly lower triangular
- Diagonally implicit:
A is lower triangular
(solve s systems of m eqns.)

Absolute Stability

$$u'(t) = \lambda u(t)$$

RK method gives:

$$Y_i = U^n + \lambda k \sum_j a_{ij} Y_j$$

$$U^{n+1} = U^n + \lambda k \sum_j b_j Y_j$$

$$Y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_s \end{bmatrix}$$

$$\mathbb{1} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \quad z = \lambda k$$

$$Y = \mathbb{1} U^n + z A Y$$

$$(I - zA)Y = \mathbb{1} U^n$$

$$Y = (I - zA)^{-1} \mathbb{1} U^n$$

$$U^{n+1} = U^n + z b^T Y$$

$$U^{n+1} = U^n + z b^T (I - zA)^{-1} \mathbb{1} U^n$$

$$U^{n+1} = R(z) U^n$$

abs. stab. function

$$R(z) = 1 + zb^T(I - zA)^{-1}\mathbf{1}$$

In general $R(z)$ is a rational function.

Recall: $\frac{1}{1-x} = 1 + x + x^2 + \dots$

So $(I - zA)^{-1} = \sum_{j=0}^{\infty} z^j A^j$

$$R(z) = 1 + z \sum_{j=0}^{\infty} z^j b^T A^j \mathbf{1}$$

Since $b^T \mathbf{1} = 1$:

$$R(z) = 1 + z + \sum_{j=1}^{\infty} z^{j+1} b^T A^j \mathbf{1}$$

If A is strictly lower triangular, $A^s = 0$

So for explicit methods:

$$R(z) = 1 + z + \sum_{j=1}^{s-1} z^{j+1} b^T A^j \mathbf{1} \quad (\text{polynomial of degree } s)$$

Absolute stability: $|R(z)| \leq 1$

Exact solution of $u'(t) = \lambda u$:

$$u(t) = e^{\lambda t} \eta$$

$$u(t_n + k) = e^{\lambda k} u(t_n)$$

So we should have $R(z) \approx e^z$

For a method of order p ,

$$\begin{aligned} R(z) &= e^z + \mathcal{O}(z^{p+1}) \\ &= \sum_{j=0}^p \frac{z^j}{j!} + \mathcal{O}(z^{p+1}) \end{aligned}$$

For example, for Kutta's method:

$$R(z) = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!}$$

These conditions are necessary for order p , but not sufficient.

For instance, for $p=3$ we also need

$$\sum_{i=1}^s b_i \left(\sum_{j=1}^s a_{ij} \right)^2 = \frac{1}{3}$$

$$u(t_{n+1}) = u(t_n) + k u'(t_n) + \frac{k^2}{2} u''(t_n) + \dots$$

$$u'_i(t) = f_i(u)$$

$$u'' = f' f$$

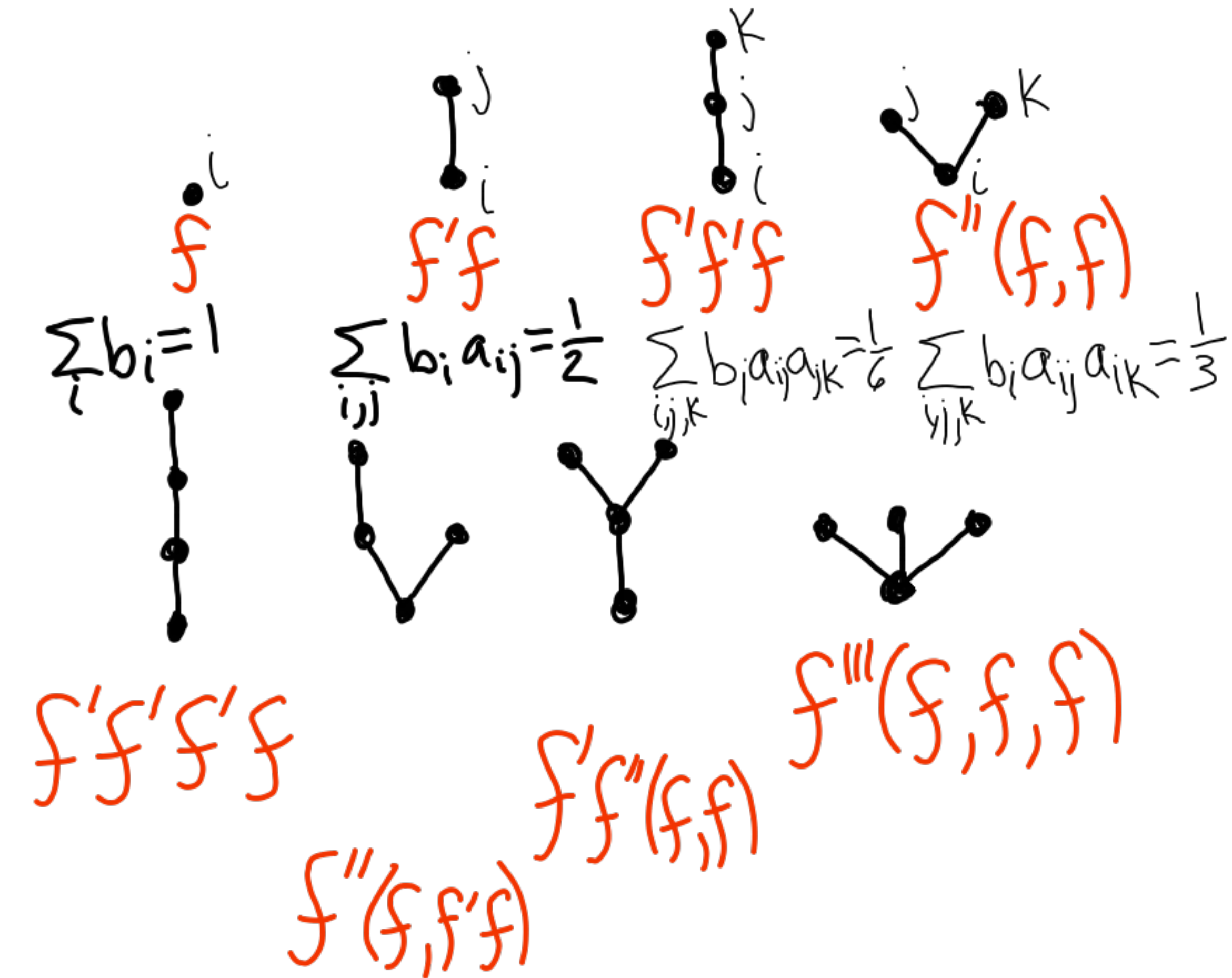
$$u''_i(t) = \sum_{j=1}^m \frac{\partial f_i}{\partial u_j} \frac{\partial u_j}{\partial t} = \sum_{j=1}^m \frac{\partial f_i}{\partial u_j} f_j$$

$$u'''_i(t) = \sum_{j=1}^m \sum_{k=1}^m \left(\frac{\partial^2 f_i}{\partial u_j \partial u_k} f_k f_j + \frac{\partial f_i}{\partial u_j} \frac{\partial f_j}{\partial u_k} f_k \right)$$

$$u''' = f''(f, f) + f' f' f$$

$$u^{(4)} = f'''(f, f, f) + 3f''(f, f'f) + f'f''(f, f) + f'f'f'f$$

There is a 1-1 correspondence between these terms and rooted trees:



Error Estimation and Step Size Control

How do we choose a step size?

- Accuracy (small local error)
- Stability (errors not amplified)

We want to use the largest step size such that

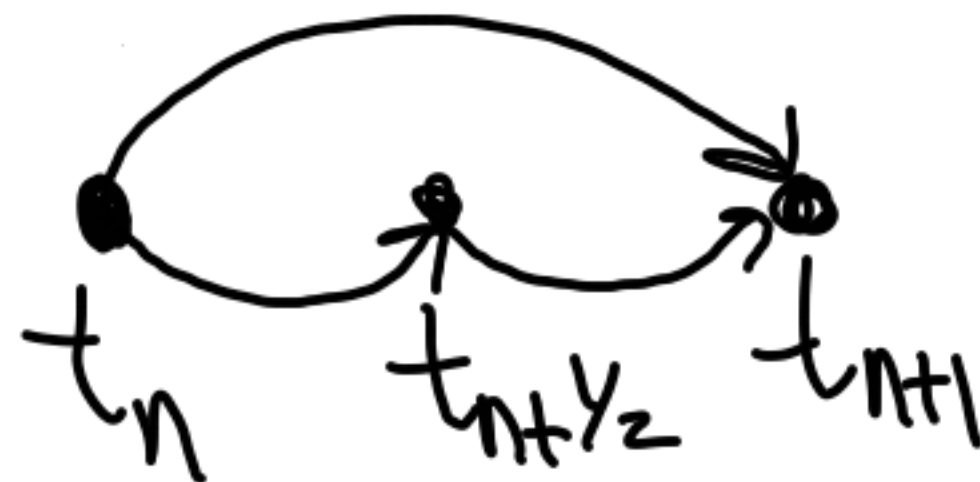
$$\tau < \varepsilon$$

local error

(depends on k)

tolerance

Richardson Extrapolation



$$U_K^{n+1} \quad U_{K/2}^{n+1}$$

$$U_K^{n+1} = u(t_{n+1}) + \tau_K^n \quad \tau_K^n = CK^p + O(K^{p+1})$$

$$U_{K/2}^{n+1} = u(t_{n+1}) + \tau_{K/2}^n \quad \tau_{K/2}^n = C\left(\frac{K}{2}\right)^p + O(K^{p+1})$$

$$U_K^{n+1} - U_{K/2}^{n+1} = CK^p \left(1 - \frac{1}{2^p}\right) = O(K^p)$$

Estimates the error (δ)

Embedded RK pairs

$$Y_i = U^n + k \sum_{j=1}^s a_{ij} f(Y_j)$$

$$U^{n+1} = U^n + k \sum_j b_j f(Y_j)$$

$$\hat{U}^{n+1} = U^n + k \sum_j \hat{b}_j f(Y_j)$$

$$\tau = \mathcal{O}(k^p)$$

$$\hat{\tau} = \mathcal{O}(k^{p-1})$$

$$\underbrace{\|U^{n+1} - \hat{U}^{n+1}\|}_{\substack{\text{Error} \\ \text{estimate} \\ \delta}} = (u(t_{n+1}) + k\tau) - (u(t_{n+1}) + k\hat{\tau})$$
$$= k|\tau - \hat{\tau}| = \mathcal{O}(k^p)$$

Step size adaptation

① Take a step of size k and compute error estimate δ

②a If $\delta > \varepsilon$, redo the step with a smaller size

②b If $\delta < \varepsilon$, continue to next step (possibly with a larger step size)